

Training the QFT operator: three findings

This is the synthesis for the `results/training/` studies. They share one subject — the parametric QFT operator as a learned image transform — and one motivating gap. On DIV2K-8q the headline random-init `qft` basis trains to 31.29 dB at $\rho = 0.20$, while `blocked_8` (a `BlockedBasis(QFTBasis(3,3), 5, 5)`, a strictly **less** expressive submanifold of `QFT(8,8)`) reaches 32.26 dB. A more expressive family ending up 0.97 dB **worse** is the puzzle. Is 31.29 a ceiling of the QFT family, or a floor of the optimiser? The three studies answer this from three directions, and a fourth study asks whether the training objective itself was well-chosen.

Finding 1 — The blocked optimum lies inside the QFT family

(`1_structure_inclusion/`) Take the trained blocked solution, embed its 48 tensor entries into a full `QFTBasis(8, 8)` with every other gate pinned to identity — a point whose operator equals the trained blocked one bit-exactly — then train all 288 parameters from there. The warm-started QFT reaches 32.26 dB at $\rho = 0.20$, **exactly the blocked optimum**, versus 31.29 dB from random init. (QuickDraw: warm-start 30.07 dB reproduces the blocked source 30.05 within 0.02 dB; random init lags 5.7 dB.) So the blocked optimum is reachable from inside `QFT(8,8)`; the headline gap is an **optimisation** failure of random-init Adam, not a structural limit of the family. **Block-QFT $\subset$ QFT**.

Finding 2 — Direct training reaches good QFT operators

(`2_direct_training/`) The analytic-QFT init is not required. Initialising every gate at its manifold identity ($T_{t=0}(x) = x$) and training under the identical budget reaches 31.66 dB at $\rho = 0.20$ — 0.37 dB **above** the analytic-init `qft`. What carries the result is the **gate topology** (which qubits the H and controlled-phase gates act on), shared by both inits, not the analytic phase values. Progressive **gate-unfreezing** — thawing one gate at a time and training each to a plateau — goes further: block-growth order from identity reaches 31.98 dB, approaching the blocked optimum, and unfreeze order matters (31.8–32.0 dB for block-growth / left→right vs 30.8 for right→left). Good QFT operators are **trainable directly**, given the right protocol.

Finding 3 — The training objective (top-k) is a real lever

(`3_training_topk/`) Every cell above optimises `MSELoss` on a fixed top 10% of coefficients while evaluation spans $\rho \in [0.05, 0.20]$. Sweeping the training top- k shows the fixed 10% is **not** optimal and the best value is dataset-dependent: DIV2K wants a **broadier** objective (top 20% reaches 31.66 dB, +0.37 over 10%), QuickDraw a **narrower** one (top 5 –10%; $\geq 15\%$ loses ≈ 0.8 dB). “Train at the rate you deploy at” is the wrong rule — one dataset-specific top- k is best across all eval rates at once.

Synthesis

The QFT family is more capable than the headline number suggests. Every route that escapes random-init Adam’s basin climbs toward the blocked optimum:

route to a QFT operator	PSNR@ $\rho = 0.20$	study
random-init joint training (headline)	31.29	baseline
identity-init joint training	31.66	Finding 2
top- $k = 20\%$ joint training	31.66	Finding 3
progressive gate-unfreezing (bg, id.)	31.98	Finding 2
warm-start from trained blocked	32.26	Finding 1
— <code>blocked_8</code> reference optimum —	32.26	structure side

Read top to bottom, 31.29 dB is an **optimisation floor**, not the family ceiling: the blocked optimum (32.26 dB) sits inside `QFT(8,8)` (Finding 1), and identity init, a broader training top- k , and progressive unfreezing each reach deeper into the family toward it (Findings 2–3) — while the training objective that produced the original floor was itself suboptimal (Finding 3). All training here is on QFT; the **structure** side (`results/structure/`) extends the comparison to the other parametric families.

Per-study detail: `1_structure_inclusion/writeup.pdf`, `2_direct_training/writeup.pdf`, `3_training_topk/writeup.pdf`.